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A Graph-Optimization-Based Tightly Coupled Multi-Source Positioning

Method for UAVs in GNSS-Denied Environments

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Abstract

To address UAV positioning in GNSS-denied environments, this study proposes a tightly coupled multi-source positioning method that integrates IMU, vision, and UWB. First, a structured visual feature optimization method is introduced to enhance feature uniformity and robustness through adaptive thresholding, density balancing, and the fusion of grayscale-gradient information. Second, a graph-optimization-based tightly coupled framework is constructed, incorporating UWB-IMU joint initialization and UWB differential residual-enhanced global constraints. By combining an adaptive sliding window with an IMU pre-integration compensation mechanism, the framework balances accuracy and real-time performance while mitigating temporal asynchrony issues. Experimental validation shows that the optimized visual features achieve an 11.8% increase in ORB matched points and a 5% improvement in accuracy compared with ORB, while demonstrating significantly better distribution uniformity than both SIFT and ORB. Finally, tests on the EuRoC dataset reveal that the proposed positioning method achieves an RMSE below 0.047 m across five sequences, with MAE reduced by 80.9% and 21.9% compared with VINS-Fusion and ORB-SLAM3, respectively. In real flight experiments, the method achieves an RMSE of 0.480 m under high-dynamic conditions and 0.286 m under steady-state conditions, validating its sub-meter accuracy and applicability in real-world environments.

Keywords: Denied Environment; Unmanned Aerial Vehicle; Positioning; Multi-Sensor Fusion

1 Introduction

In recent years, unmanned aerial vehicles (UAVs) have seen widespread application in military reconnaissance, disaster relief, and other fields. The success of their autonomous flight and mission execution is contingent upon the utilization of precise and reliable positioning systems. Presently, Global Navigation Satellite Systems (GNSS) function as the predominant positioning methodology for Unmanned Aerial Vehicles (UAVs). However, in complex electromagnetic environments or adversarial scenarios, GNSS signals are highly susceptible to blockage, interference, or spoofing attacks. This hinders UAVs from acquiring precise positional data, jeopardizing flight safety and exerting a substantial negative influence on mission success rates. Consequently, achieving high-precision positioning for UAVs in GNSS-denied environments has become a critical technology and research focus for ensuring safe flight operations and mission success.

Visual localization has been demonstrated to be an effective positioning method for UAVs in GNSS-denied environments. For instance, SLAM facilitates high-precision positioning in small-

scale, uncharted environments [1-3], while image matching-based localization is predominantly employed for long-range, wide-view scenarios. This approach utilizes a global estimation method that involves the comparison of real-time imagery with terrain databases [4-5]. The accuracy of visual positioning is contingent upon the precision of feature matching and relative pose estimation between successive frames in the front-end processing. This is imperative for the suppression of positioning drift. Common front-end methods include feature point-based approaches, optical flow-based approaches, direct methods, and deep learning-based methods [6-8]. Optical flow methods are predicated on the assumption of constant brightness, thereby enabling the tracking of pixel motion. For instance, SVO2.0 [9-10] integrates sparse direct methods with local optimization to enhance stability in large-scale scenes. However, both the optical flow and direct methods are sensitive to luminance variations and are prone to failure due to noise, blurring, and abrupt lighting changes. Deep learning-based approaches, exemplified by DeepVO[11] and TartanVO[12], employ convolutional neural networks (CNNs) to learn motion from image sequences. However, these approaches are heavily reliant on substantial datasets and substantial computational resources, which compromises their effectiveness in real-time operations on unmanned aerial vehicles (UAVs). Consequently, feature-based methods remain the predominant front-end approach for UAV visual localization.

Feature point methods are a class of algorithms that estimate the motion state of a carrier by extracting feature points from consecutive frames and matching them. This methodology has been demonstrated to extract feature points that are resistant to rotation and translation, thereby ensuring a high degree of robustness. Among the various visual localization algorithms grounded in feature point methods, ORB-SLAM [13] is a noteworthy exemplar. This system establishes a comprehensive monocular real-time localization framework by integrating loop detection and graph optimization techniques based on ORB features. Its successor, ORB-SLAM2 [14], extends support to stereo and RGB-D inputs, further broadening the system's applicability. However, the mission areas of drone operations frequently encompass low-texture scenes, such as sandy terrains, fields, and water surfaces. This phenomenon results in the presence of sparse feature points, which, in turn, leads to challenges in matching. Consequently, the continuity and robustness of trajectory estimation are severely compromised. Concurrently, the coexistence of bright skies and shadowed surfaces in images engenders local grayscale distortion, thereby significantly degrading the matching accuracy of descriptors based on grayscale difference encoding. Therefore, enhancing feature perception capabilities in low-texture environments and improving descriptor robustness under dynamic lighting conditions are critical for achieving high-precision visual positioning in UAVs.

Given the visual-only positioning's inability to meet high-precision requirements under highly dynamic conditions, multi-source fusion positioning is currently the preferred approach. Inertial measurement units (IMUs) are capable of providing high-frequency, continuous attitude

information. Consequently, extensive research has focused on integrating IMU data with visual information to enhance positioning system accuracy. For instance, the VINS-Mono system [15] employs a tightly coupled optimization strategy to jointly model visual feature observations with IMU pre-integration measurements, achieving high-precision pose estimation. ORB-SLAM3[16] incorporates IMU observations by extending the original visual SLAM framework to support simultaneous visual-inertial mapping and localization. In addition to the integration of visual-IMU combinations, significant advancements have been made in recent years in the field of UWB ranging information integration. The VIU-Net system[^]([17]) employs deep learning methods to fuse visual, IMU, and UWB data, thereby enhancing the accuracy of UAV positioning. The RFG-TVIU system [18] employs factor graph optimization to dynamically adjust observation weights without requiring prior calibration of UWB base station positions, effectively addressing positioning challenges in complex environments. Consequently, the integration of information from visual, IMU, and UWB data is of paramount importance for enhancing the positioning accuracy of UAVs in complex, dynamic environments.

The present paper endeavors to address the challenge of high-precision positioning for UAVs in GNSS-denied environments. To this end, a multi-source positioning method has been designed that integrates IMU, visual, and UWB information. For the visual component, the proposal calls for an optimized structured visual feature approach that effectively mitigates the susceptibility of visual observations to low-texture repetitive structures and dynamic lighting variations. In light of the substantial observational discrepancies among IMU, visual, and UWB sensors, this paper puts forth a graph-optimization-based positioning method that integrates IMU/visual/UWB data. This method unifies the integration of visual reprojection residuals, IMU pre-integration, and UWB ranging information. Within the framework of a factor graph optimization model, a joint optimization model incorporating multi-source residual terms is constructed to achieve tightly coupled state estimation of visual, inertial, and ranging information.

2 Structured Visual Feature Optimization Method

2.1 Structured Adaptive FAST Feature Extractor

This paper proposes a structured, adaptive FAST feature extractor based on the ORB feature extraction module. It enhances the quality of feature point extraction and the spatial distribution balance in complex images by introducing local adaptive thresholds, feature point density equalization, and edge suppression in distorted regions. Fig. 1 illustrates the extraction process.

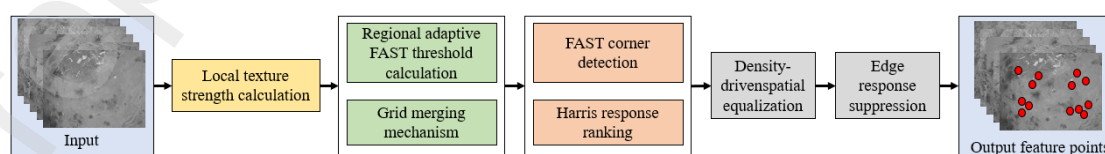


Fig.1 Structured Adaptive FAST Feature Extraction Flowchart

(1) An adaptive thresholding mechanism for local texture perception.

Divide the image into fixed-size grid regions, typically using a uniform division of 16×16 or 32×32 . Then, calculate the standard deviation of gradient intensity within each grid to measure texture complexity. Let the set of pixels in grid $G_{i,j}$ be $\Omega_{i,j}$. Then, the gradient standard deviation for this region is defined as follows:

$$\sigma_{i,j} = \sqrt{\frac{1}{|\Omega_{i,j}|} \sum_{(x,y) \in \Omega_{i,j}} (\|\nabla I(x,y)\| - \|\overline{\nabla I}\|_{i,j})^2} \quad (1)$$

Here, $\|\nabla I(x,y)\|$ represents the magnitude of the pixel gradient, and $\|\overline{\nabla I}\|_{i,j}$ denotes the average gradient magnitude for the grid. To enable each grid to respond to its local texture variation, a region-adaptive FAST threshold is constructed.

$$T_{i,j}^{FAST} = T_0 \cdot \left(1 - \frac{\sigma_{\max} - \sigma_{i,j}}{\sigma_{\max}}\right) \quad (2)$$

T_0 represents the global initial threshold, $\sigma_{\max}$ represents the maximum texture standard deviation across all grids, and $\sigma_{i,j}$ represents the threshold value. Lower complexity results in a smaller corner detection threshold, while higher complexity yields a larger threshold.

If the texture standard deviations of adjacent grids $G_{i,j}$ and $G_{i+1,j}$ are $\sigma_{i,j}$ and $\sigma_{i+1,j}$ respectively, the texture complexity difference metric between them is:

$$\delta_{\sigma} = \frac{|\sigma_{i,j} - \sigma_{i+1,j}|}{\max(\sigma_{i,j}, \sigma_{i+1,j})} \quad (3)$$

If $\delta_{\sigma} < \delta_{merge}$, the texture structures are considered similar and can be merged into a shared region $G_{i \cup i+1,j}$. This paper adopts an empirical value $\delta_{merge} = 0.1$. The FAST detection threshold for the merged region is calculated as follows:

$$T_{i \cup i+1,j}^{FAST} = \frac{T_{i,j}^{FAST} + T_{i+1,j}^{FAST}}{2} \quad (4)$$

(2) Feature point density-driven spatial equalization

This method builds upon the quad-tree strategy [21][22] by introducing a feature-point density-driven spatial partitioning mechanism. It identifies potential cluster areas and prioritizes spatial partitioning in these regions by analyzing the average density of corner points and the geometric

dispersion within local regions.

Let the image be partitioned into region R , which contains n vertices. The average distance between vertices is $\bar{p}$, and the variance of the distances is:

$$\delta_d^2 = \frac{1}{n} \sum_{i=1}^n \|p_i - \bar{p}\|^2 \quad (5)$$

When the value of δ_d^2 exceeds the set threshold of δ_{th} , it is determined that the feature points within the region exhibit a non-uniform distribution or dense clustering. This necessitates a quadtree split. The splitting process is as follows:

- ① Divide it into four subregions if the number of corner points in the current region exceeds the preset threshold N_{split} , or if $\delta_d^2 > \delta_{th}$.
- ② Terminate the split if the current tree node depth has reached maximum depth d_{max} or if the density difference between the corner points of the subintervals is less than 15%.
- ③ To ensure both quantity control and quality priority, each subregion retains only the top K corner points with the highest response values.

(3) Edge response suppression method

This paper proposes an edge response suppression mechanism to address image edge distortion caused by wide-angle lenses on drones. This mechanism eliminates interference from false corner points during matching. Denote the image center as $c = (x_c, y_c)$ and a corner point location as $p = (x, y)$. The edge suppression weight function is defined as follows:

$$w(p) = \exp(-\alpha \cdot \|p - c\|^2) \quad (6)$$

In this context, α controls the rate of edge attenuation, which ranges from:

$$\alpha = \frac{1}{(0.4 \cdot W)^2} \quad (7)$$

W represents the width of the image. The "reliable response value" for the final corner is calculated as follows:

$$R_{final}(p) = R_{Harris}(p) \cdot w(p) \quad (8)$$

2.2 Gradient-Gray Dual-Modality G²-ORB Descriptor

This paper proposes the G²-ORB (Gradient-Gray ORB) descriptor, which integrates gradient direction and gray-scale intensity, as shown in Fig. 2, building upon traditional BRIEF[23]-[25].

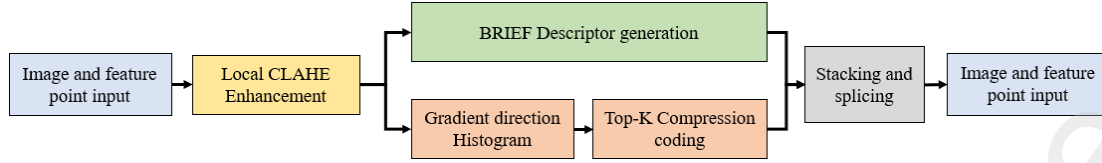


Fig. 2 G²-ORB Descriptor Generation Flowchart

(1) Light-normalized processing

First, perform Contrast-Limited Adaptive Histogram Equalization (CLAHE) processing on the $I_N(x, y)$ image block in this window to obtain the normalized $\hat{I}_N(x, y)$ image.

(2) BRIEF foundation layer construction

The BRIEF method is employed on the normalized image to generate a 256-bit binary descriptor. Two hundred fifty-six pairs of pixels are sampled around the feature point, and a basic descriptor vector is generated through grayscale comparison.

$$d_{BRIEF} = [\tau_1, \tau_2, \dots, \tau_{256}] \in \{0, 1\}^{256} \quad (9)$$

(3) Directional layer enhancement design

G²-ORB incorporates gradient direction histogram coding to enhance structural direction representation capabilities, while reusing existing gradient components $I_x(x, y)$ and $I_y(x, y)$ to avoid redundant computations.

Using the obtained gradient information, calculate the gradient direction of the pixel.

$$\theta(x, y) = \arctan\left(\frac{I_y(x, y)}{I_x(x, y)}\right) \quad (10)$$

The azimuth angle range is $[0^\circ, 180^\circ)$ and is divided into eight equal bins. The frequency with which gradient directions fall into each bin is counted to construct a directional histogram.

$$g_j = \sum_{(x, y) \in N(p)} w(x, y) \cdot \delta(\theta(x, y) \in bin_j, j = 1, 2, \dots, N) \quad (11)$$

Among these, $w(x, y)$ represents the pixel gradient magnitude and $\delta(\bullet)$ denotes the logical indicator function. The directional histogram is normalized and quantized to compress the dynamic range, enhance contrast, and improve numerical stability. This process ultimately generates a direction-enhanced vector.

$$g = [g_1, g_2, \dots, g_8] \in \mathbb{N}^8 \quad (12)$$

(4) Sparse coding and descriptor concatenation

Direction histograms are floating-point/integer vectors, and direct matching requires calculating L1/L2 distances, making it difficult to meet real-time embedded requirements. Therefore, a multi-dominant-direction hard-coding strategy is adopted to achieve sparse compression encoding.

From the normalized histogram, the bin indices $\{i_1, i_2, \dots, i_k\}$ corresponding to the Top-K

maximum response values are selected to construct one-hot encoded values:

$$code = \sum_{k=1}^K 2^{i_k} \quad (13)$$

This encoding requires only one byte for representation. It exhibits excellent sparsity and interpretability. It can also be directly appended to the end of the BRIEF descriptor to form the final output.

$$\mathbf{d}_{G^2-ORB} = [\mathbf{d}_{BRIEF}, code] \in \{0,1\}^{256+N} \quad (14)$$

During the matching process, the BRIEF component uses fast Hamming distance comparisons, while the directional encoding component uses lookup tables to determine directional consistency, balancing efficiency and discriminative power.

3 A Graph Optimization-Based Positioning Method Integrating IMU, Vision, and UWB

The factor graph-optimized multi-source fusion positioning method proposed in this paper integrates IMU, visual, and UWB observation information. By uniformly constructing visual reprojection residuals [26][27], IMU pre-integration residuals [28][29], and UWB ranging residuals [30][31], this method establishes a tightly coupled joint optimization model within the factor graph framework. This enables highly consistent state estimation of visual, inertial, and ranging information. The overall method architecture is illustrated in Fig. 3.

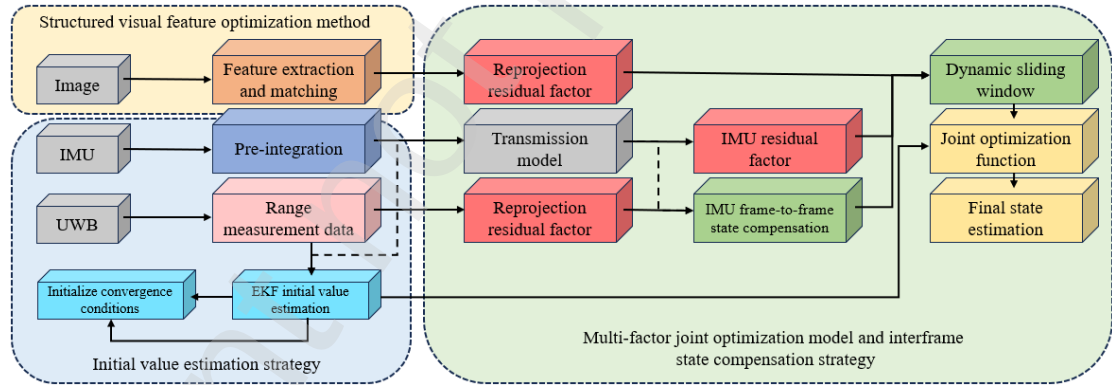


Fig. 3 Architecture Diagram of the Fusion Positioning Method.

3.1 Integrating Positioning Theory and Modeling

The spatial relationships among various observation data from multi-source fusion positioning of UAVs in different scenarios are illustrated in Fig. 4.

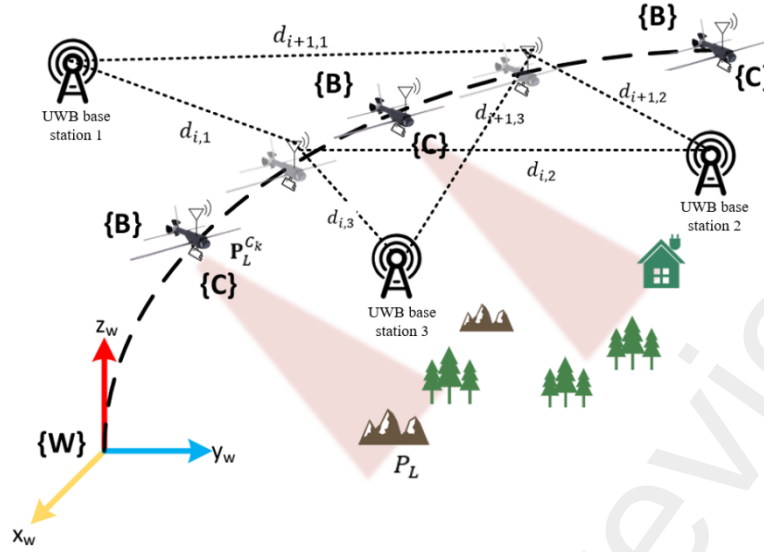


Fig. 4 Schematic Diagram of Spatial Relationships Among Observation Sources in Fusion Positioning.

This paper uses the global reference frame $\{W\}$ as the world coordinate system. The motion state of the UAV is represented in the carrier coordinate system $\{B\}$, while visual observations are based on the camera coordinate system $\{C\}$. The rigid transformation between the camera and UAV coordinate systems is obtained through external parameter calibration. The IMU provides continuous state predictions; the UWB acquires distance observations between the UAV and base stations (e.g., $d_{i,1}, d_{i,2}$); and the visual module constructs reprojection residuals by matching keypoints between frames. These three types of observations provide complementary spatiotemporal information and form the foundation of data and constraints for graph optimization.

As shown in Fig. 5, the system unifies the conversion of sensor data to the ENU coordinate system through multiple coordinate systems. The final output is converted to geographic coordinates for visualization and track reconstruction. This provides a unified reference framework for subsequent multi-source modeling.

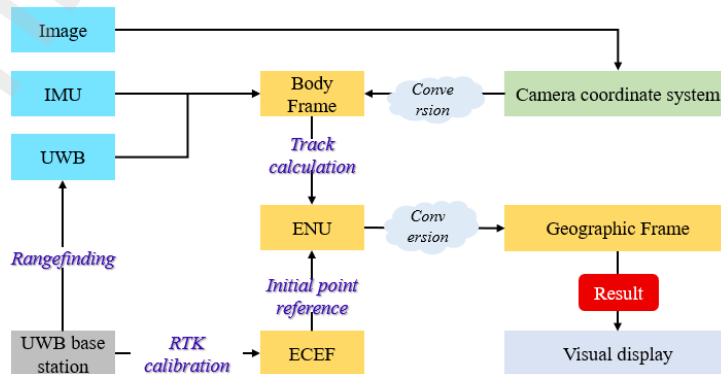


Fig. 5 Coordinate System Transformation for a Multi-source Fusion Positioning System

3.2 Optimized Factor Diagram Modeling Method Under Multi-Source Observation Constraints

Factor graphs, which serve as graphical representations of optimization, express dependencies among state variables and residual constraints from sensor observations. They offer excellent scalability and interpretability. Fig. 6 shows the factor graph structure used in this paper, in which system states are represented by variable nodes. Adjacent states are connected via IMU pre-integration factors, forming continuous temporal constraints. The UWB ranging factor and visual reprojection factor serve as external observation constraints, providing global positioning and reprojection information, respectively. The UWB factor calculates residuals based on the Euclidean distance between the current frame and multiple base stations. This achieves fusion by integrating ranging observations with weights. However, due to temporal offsets between ranging data and visual keyframes, ranging data is usually associated with the nearest keyframe.

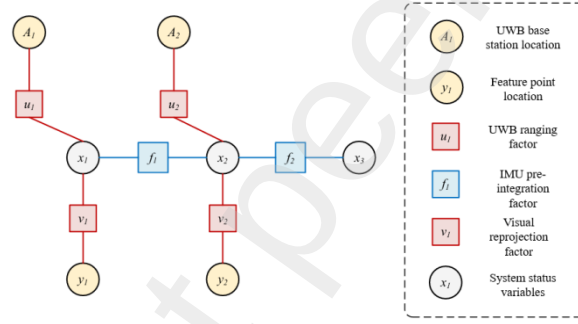


Fig. 6 Schematic Diagram of The Factor Diagram Structure.

3.2.1 Initial State Estimation Strategy for UWB-IMU Integrated Systems

During the initial phase of drone operations, systems are susceptible to significant motion or observation gaps. Relying solely on inertial measurement units (IMUs) can lead to cumulative errors due to zero bias. This paper uses an extended Kalman filter (EKF) to combine UWB and IMU data. Pre-optimizing the fusion of multi-base station ranging measurements with IMU data provides reliable initial values for subsequent sliding window optimization.

$$x_0(t) = [p_0(t), v_0(t), q_0(t), b_a(t), b_g(t)] \quad (15)$$

The system's state model is shown in Eq. (16), and the state transition model is shown in Eq. (17).

$$x(t) = [p(t), v(t), q(t), b_a(t), b_g(t)] \quad (16)$$

$$\begin{cases} \dot{p}(t) = v(t) \\ \dot{v}(t) = R(t) \bullet (a_m(t) - b_a(t) - n_a(t)) + g \\ \dot{q}(t) = \frac{1}{2} \Omega(w_m(t) - b_g(t) - n_g(t)) \bullet q(t) \\ \dot{b}_a(t) = n_{ba}(t), \quad \dot{b}_g(t) = n_{bg}(t) \end{cases} \quad (17)$$

Among these, $p(t) \in \mathbb{R}^3$ represents position, $v(t) \in \mathbb{R}^3$ represents velocity, $q(t) \in H$ represents attitude, and $b_a(t), b_g(t)$ represents zero bias. $\Omega(\bullet)$ represents the 4×4 matrix generated by left-multiplying quaternions, and $n_{ba}(t), n_{bg}(t)$ represents random walk noise. During the prediction phase, the system relies solely on IMU measurements for state propagation using the following prediction model:

$$\begin{cases} x_{i+1}^- = f(x_i, \hat{a}_i, \hat{w}_i) \\ P_{i+1}^- = F_i P_i F_i^T + Q_i \end{cases} \quad (18)$$

Among these, $\hat{a}_i, \hat{w}_i$ represents the acceleration and angular velocity observed by the IMU, and Q_i denotes the process noise covariance matrix. Linearizing the state transition function at the current state point yields the state transition Jacobian matrix, F_i .

$$F_i = \left. \frac{\delta f}{\delta x_i} \right|_{\hat{x}_i} = \begin{pmatrix} I & I\Delta t & 0 & 0 & 0 \\ 0 & I & -R_i \lfloor \hat{a}_i \rfloor_{\times} \Delta t & -R_i \Delta t & 0 \\ 0 & 0 & I - \lfloor \hat{w}_i - b_g \rfloor_{\times} \Delta t & 0 & -I\Delta t \\ 0 & 0 & 0 & I & 0 \\ 0 & 0 & 0 & 0 & I \end{pmatrix} \quad (19)$$

During the update phase, UWB ranging observations are introduced to correct the cumulative error generated by IMU integration. The observation model is shown in Eq. (20).

$$d_{ij} = \|p_i - p_j\| + n_{ij} \quad (20)$$

The UWB observation model can be simplified into a nonlinear Euclidean distance constraint related to the position variable. The Jacobian of this constraint with respect to the position state is:

$$H_{ij} = \frac{\partial d_{ij}}{\partial p_i} = \frac{p_i - p_j}{\|p_i - p_j\|} \quad (21)$$

This allows for the development of an extended Kalman filter update process that includes Kalman gain calculation, state correction, and covariance reduction.

$$K_{ij} = P_i^- H_{ij}^\top (H_{ij} P_i^- H_{ij}^\top + R_j)^{-1} \quad (22)$$

$$x_i^+ = x_i^- + K_{ij} (d_{ij} - \hat{d}_{ij}) \quad (23)$$

$$P_i^+ = (I - K_{ij} H_{ij}) P_i^- \quad (24)$$

This paper uses a sliding window strategy to estimate and fuse initial conditions. During the initialization phase, static initialization is first performed based on multi-frame unwrapping. Then, velocity and attitude estimation is performed via IMU integration. Finally, keyframe states are progressively constructed. Convergence of the initial state estimation requires satisfying two sets of conditions. First, state convergence requires accumulating at least ten valid observations within the sliding window to achieve attitude estimation convergence (with the quaternion model length approaching one and an error below 0.05), maintaining stable ranging residuals within 0.2 m for five consecutive frames, and stabilizing the IMU bias. Second, visual activation requires the successful detection of at least one pair of co-viewed keyframes with stable feature matching counts that exceed the threshold. Once these conditions are met, the system transitions to backend graph optimization. The graph structure is constructed from the initial state, and inertial, odometry, and visual factors are progressively incorporated to achieve tri-modal joint estimation.

3.2.2 Multi-factor Combined Optimization Model Construction

This paper proposes a tightly coupled sliding window optimization model that integrates IMU, visual, and UWB information. The model is based on a factor graph and represents states and observations as nodes and factors, respectively. It achieves intra-window state estimation through nonlinear optimization.

The set of optimization variables is constituted by all states within the sliding window.

$$\mathbf{X} = \{x_0, x_1, \dots, x_n\} \quad (25)$$

Here, $\mathbf{X}$ denotes the set of all state variables within the sliding window, and x_i represents the state variables of the i node. At frame i , observations are acquired from the IMU, visual, and UWB sensors, respectively. Corresponding factor constraints are then established, as shown in Eq. (26), (27), and (28).

$$\min_{\{T_i, P_j\}} \sum_{i,j} \|z_{ij} - \pi(T_i \bullet P_j)\|^2 \quad (26)$$

$$\min_{\mathbf{X}} \sum_{(i,j)} r_{IMU,ij}^\top \sum_{ij}^{-1} r_{IMU,ij} \quad (27)$$

$$\min_X \sum_i \sum_{j=1}^N \rho(r_{UWB,ij}^2) \quad (28)$$

This paper proposes a composite ranging factor that integrates absolute and differential ranging residuals. This enhancement improves the global observation capability of UWB and suppresses errors. The result is an observation model with both absolute and relative constraints. The absolute ranging residual is modeled as follows:

$$r_{abs,ij} = d_{ij} - \|p_i - p_j\| \quad (29)$$

where d_{ij} represents the UWB ranging observation between the system at time i and the j th base station, p_i denotes the system position to be optimized, and p_j indicates the known base station positions. Furthermore, differential ranging residual modeling is introduced, expressed as:

$$r_{diff,ij} = (d_{i+1,j} - d_{i,j}) - (\|p_{i+1} - p_j\| - \|p_i - p_j\|) \quad (30)$$

Ultimately, the constructed UWB composite factor can be expressed as:

$$r_{UWB}^{(i)} = \begin{bmatrix} r_{abs,ij} \\ r_{diff,ij} \end{bmatrix} \quad (31)$$

During image optimization, each frame state node establishes a residual chain with multiple UWB base stations, forming a stable global and local constraint structure. The UWB factor, along with the IMU pre-integration residual and visual reprojection residual, constitutes the system's overall optimization objective function. This function is expressed as follows:

$$X^* = \arg \min_X \{ \sum_k \rho(\|r_{vision,k}\|^2) + \sum_l \rho(\|r_{IMU,l}\|^2) + \sum_m \sum_j \rho(\|r_{IMU,mj}\|^2) \} \quad (32)$$

$\rho(\cdot)$ denotes a robust kernel function. This paper adopts the Huber kernel, defined as follows:

$$\rho(s) = \begin{cases} s^2, & s \leq \delta \\ 2\delta\sqrt{s} - \delta^2, & s > \delta^2 \end{cases} \quad (33)$$

To achieve robust residual-weighted modeling during global optimization, different turning thresholds are set for visual, UWB, and velocity residuals. During each iteration of the sliding window update, the system builds a complete factor graph structure and performs multiple rounds of nonlinear optimization. Ultimately, it outputs the optimal state estimate within the current window.

3.2.3 Dynamic sliding window policy and marginalization mechanism

The fundamental strategy of sliding window optimization is to maintain a set of states based on the key frame states from the most recent N frames.

$$X_{win} = \{x_{t-N+1}, \dots, x_t\} \quad (34)$$

Among these, x_t represents the current frame state and x_{t-N+1} represents the earliest frame in the window. A fixed window length cannot accommodate the varying requirements of unmanned aerial vehicles across different mission phases. To address this issue, this paper proposes an adaptive window adjustment strategy based on motion intensity. By analyzing velocity change intensity, the window frame count N is dynamically adjusted. The following motion intensity indicators are defined:

$$S = \frac{1}{N-1} \sum_{i=t-N+1}^{t-1} \|v_{i+1} - v_i\| \quad (35)$$

If $S > \varepsilon_{high}$, increase the frame rate to enhance observational redundancy; otherwise, reduce the frame rate to lower computational load. To prevent frequent window fluctuations, upper and lower limits are set for the frame rate during dynamic adjustment.

$$N_{min} \leq N \leq N_{max} \quad (36)$$

Achieve a balanced trade-off between real-time performance, accuracy, and stability by setting $N_{min}=7$ and $N_{max}=12$.

When the sliding window updates, the earliest frame's state is removed. This information is converted into a prior residual through marginalization and continues to participate in subsequent optimization, thereby preserving the influence of historical observations.

Suppose the residual function in the current graph can be decomposed into two categories:

$$r(X) = \begin{bmatrix} r_m(x_m, x_r) \\ r_o(x_r) \end{bmatrix} \quad (37)$$

Among these, x_m represents the old state to be marginalized, while x_r denotes the retained state within the sliding window; r_m represents observational residuals associated with x_m , while r_o denotes other residuals solely related to x_m . The removed state variable x_m is not discarded outright; instead, its influence is transformed into prior information through marginalization and

reintroduced as residuals into the graph to constrain the remaining state variable x_m . First-order linearization of the residual function at the current state point yields the following incremental equation system:

$$\begin{bmatrix} H_{mm} & H_{mr} \\ H_{rm} & H_{rr} \end{bmatrix} \begin{bmatrix} \delta x_m \\ \delta x_r \end{bmatrix} = \begin{bmatrix} b_m \\ b_r \end{bmatrix} \quad (38)$$

Among these, H_{mm} and H_{rr} represent the Heisenberg matrices for the marginalized and retained states, respectively. H_{mr} and H_{rm} denote the cross-second-order information, while b_m and b_r correspond to the linearized gradient vector terms. Eliminating δx_m using Schur complements compresses the above equations into equivalent prior information terms for the retained variables.

$$H_{\text{marg}} \delta x_r = b_{\text{marg}} \quad (39)$$

$$H_{\text{marg}} = H_{rr} - H_{rm} H_{mm}^{-1} H_{mr} \quad (40)$$

$$b_{\text{marg}} = b_r - H_{rm} H_{mm}^{-1} b_m \quad (41)$$

Perform a Cholesky decomposition on H_{marg} to construct the following equivalent prior residual:

$$r_{\text{prior}} = H_p \delta x_r - r_p \quad (42)$$

Among these, H_p represents the Cholesky decomposition of the marginalized information matrix

$H_{\text{marg}} \cdot r_p$ denotes the compressed residual term and r_{prior} is reintroduced into the optimization graph as a new marginalization factor. In summary, the objective function for the overall system optimization is:

$$X^* = \arg \min_X \left\{ \sum_k \rho(\|r_{\text{vision},k}\|^2) + \sum_l \rho(\|r_{\text{IMU},l}\|^2) + \sum_m \sum_j^N \rho(\|r_{\text{IMU},ml}\|^2) + r_{\text{prior}} \right\} \quad (43)$$

3.2.4 Dynamic Sliding Window Policy and Marginalization Mechanism

This paper proposes an inter-frame state compensation strategy that uses IMU pre-integration. It achieves spatiotemporal alignment for UWB ranging without introducing additional nodes by leveraging high-frequency IMU observations. The strategy estimates state increments through pre-integration and incorporates them as compensation terms into the UWB residual function. This approach uniformly models asynchronous observations at key frames. This approach preserves the

information constraint capability of UWB fully while maintaining the sparsity of the factor graph and avoiding graph expansion. Fig. 7 illustrates the resulting graph structure evolution.

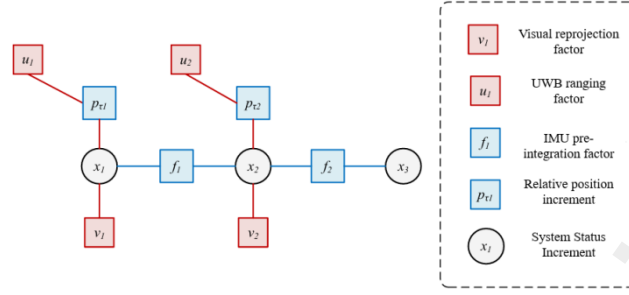


Fig.7 Factor Diagram Structure for Inter-Frame State Compensation

Set two adjacent keyframes, x_i and x_{i+1} , which correspond to times, t_i and t_{i+1} , respectively. The UWB ranging observation occurs at time $t_k^{UWB} \in (t_i, t_{i+1})$. Time alignment is completed by computing state propagation from the UWB sampling time to the keyframes via IMU pre-accumulation segment interpolation.

For the ranging observation d_k^{UWB} , using the nearest visual keyframe x_{i+1} as the alignment node, the relative position increment $\Delta p_{k \rightarrow i+1}$ is estimated based on IMU information within $[t_k^{UWB}, t_{i+1}]$. Subsequently, the compensated position is projected onto the keyframe node, with the residual function expressed as:

$$r_{UWB, i+1} = d_k^{UWB} - \|p_{i+1} + \Delta p_{k \rightarrow i+1} - p_j\| \quad (44)$$

Among these, p_{i+1} denotes the key frame position, p_j represents the base station position, and $\Delta p_{k \rightarrow i+1}$ indicates the displacement estimation from t_k^{UWB} to t_{i+1} . The position increment is calculated using “interpolative integration” approximated via the IMU pre-integration model.

$$\Delta p_{k \rightarrow i} \approx \int_{t_k^{UWB}}^{t_i} v(t) dt \quad (45)$$

Among these, $v(t)$ represents the instantaneous velocity obtained through IMU integration.

This alignment mechanism uses IMU pre-integration to bind all asynchronous UWB observations to key frames, enabling joint optimization under a unified state. The interpolation process only occurs during residual construction and introduces no new variables or additional optimization dimensions. Compared to methods that add nodes for each observation, this approach significantly reduces graph complexity while maintaining observation utilization. This enhances optimization efficiency and convergence stability.

4 Experient

This paper presents a multi-source fusion positioning method for unmanned aerial vehicles (UAVs) that integrates an inertial measurement unit (IMU), visual sensors and an ultra-wideband (UWB) ranging module. Its overall architecture is shown in Fig. 8. The system uses a Xilinx Zynq UltraScale+ MPSoC XCZU5EV as its main computing platform. The IMU uses the ADIS16505-2BMLZ model; the visual module uses the MER2-502-79U3M industrial camera; and the UWB component uses a customised ranging module and base station system. Key parameters are detailed in Table 1.

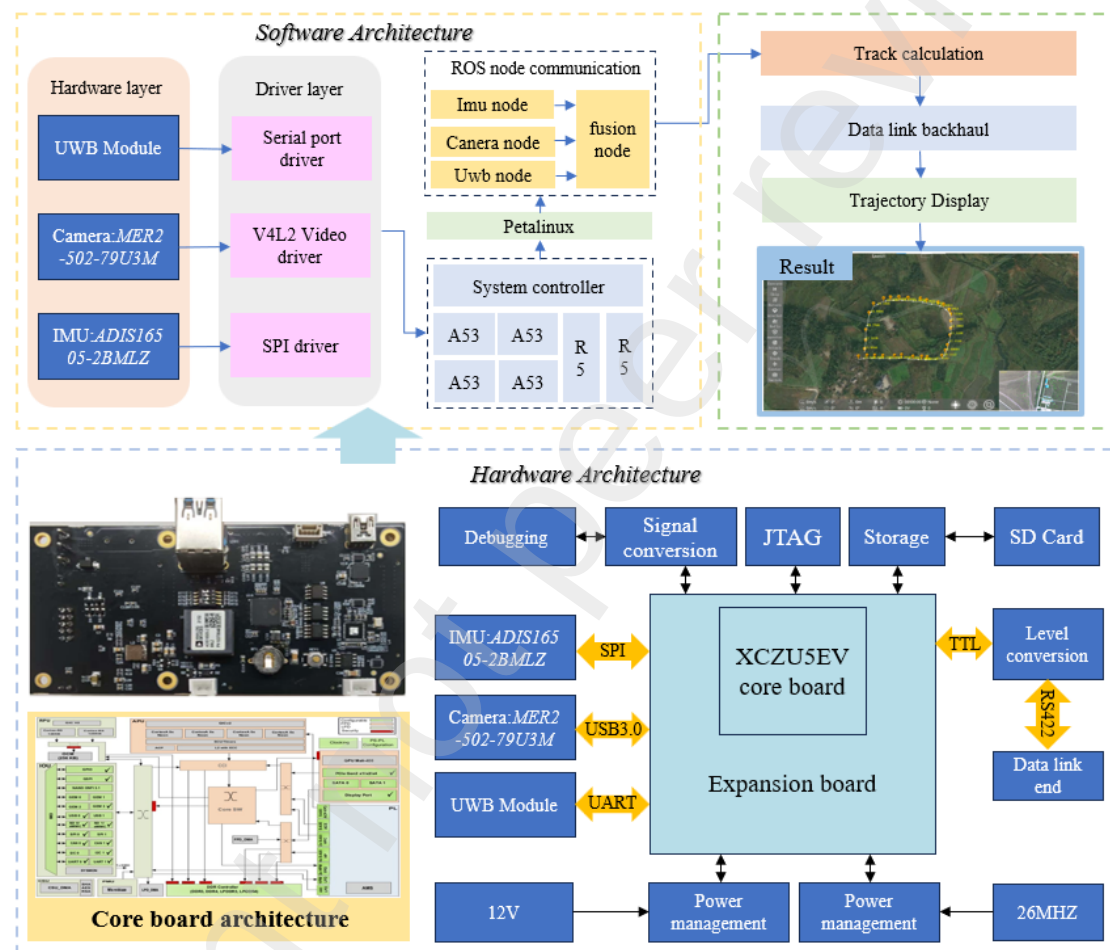


Fig. 8 Architecture of a Multi-Source Fusion Positioning System for UAVs in GNSS-Denied Environments

Table 1 UWB module performance specifications

Indicator	Parameters
Operating frequency band	3~4GHz
Bandwidth	$\geq 500\text{MHz}$
Power gain	20~30dB
Working distance	>10km
Volume	120×60×30mm

To evaluate the performance of the improved feature extraction and description method comprehensively, two types of experiment were designed. One uses typical flight image sequences captured by actual UAVs. These sequences cover common scenarios in low-altitude missions, such as lighting variations, motion blur and sparse textures. This approach accurately reflects the distribution of real-world perceptual features, demonstrating its practicality and representativeness. The other experiment uses the publicly available HPatches dataset [37-39] for validation.

Experiments were conducted using the EuRoC public dataset [40] to validate the effectiveness of the proposed IMU/vision/UWB fusion graph optimization localization method and to compare it with the VINS-Fusion [41] and ORB-SLAM3 [42] systems. The EuRoC dataset contains diverse flight sequences under various dynamic conditions and lighting scenarios. It provides high-frequency IMU and image data, which is ideal for evaluating the accuracy of vision-inertial systems. Since the dataset lacks real UWB data, this study generated simulated UWB observations based on real trajectories. Three virtual base stations were uniformly placed in the scene, and ranging was simulated by calculating the Euclidean distance between the ground truth and the base stations. Gaussian noise $\mathcal{N}(0, 0.052)$ was introduced to match actual UWB measurement errors.

This study evaluated the positioning accuracy of the proposed multi-source fusion positioning method in actual flight scenarios. To do so, the system shown in Figure 8 was mounted onto a drone platform, flight trajectories were generated, and a comparative analysis was conducted with the actual trajectories. To acquire real trajectories, the drone platform was equipped with an additional ZED-F9P high-precision GNSS positioning module.

5 Results and Discussion

5.1 Verification of Mission Image Sequences

To validate the practical adaptability and robustness of the method, a sequence of typical flight images captured by a drone platform was selected as test data. This data set included challenges such as motion blur, low texture, and lighting variations. Feature extraction and bidirectional KNN matching were performed every 20 frames. The exact registration matrix from image registration served as the ground truth. We evaluated matching accuracy by measuring the interior point rate, reprojection error, and repeatability rate. All algorithms were implemented using BFMatcher at a uniform scale. The results are presented in Table 2.

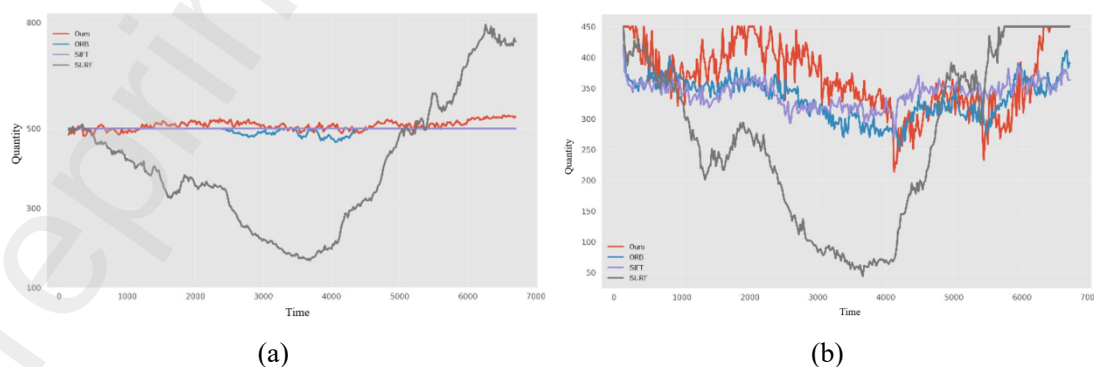
Table 2 Comparison of Experimental Results for UAV Mission Image Sequences

Method	Number of Feature Points	Number of Matching Points	In-point rate (%)	Unifor- mity	Descriptor Bias	Descriptor Information Entropy	Duplication Rate (%)	Total Duration (ms)
SIFT	500.10	340.66	98	0.45	-	0.6	83	103.29
SURF	396.01	290.76	96	0.41	-	0.71	79	31.15
ORB	499.54	332.41	88	0.35	0.51	0.98	79	16.77

Ours	500.5	371.65	93	0.54	0.49	0.98	85	20.56
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In terms of matching accuracy, the proposed method achieves an interior point rate of 93%. This rate outperforms those of ORB (88%) and SURF (96%), approaching the rate of SIFT (98%). With comparable feature point counts, the average number of matched points increases by 11.8% compared to ORB. In terms of efficiency, the runtime is 20.56 ms, which is slightly higher than ORB (16.77 ms), yet significantly lower than SURF (31.15 ms) and SIFT (103.29 ms). This achieves an excellent trade-off between accuracy and efficiency. Regarding feature distribution, the uniformity index is 0.54, which is significantly higher than the index for ORB (0.35) and other algorithms. This demonstrates how the structured extraction strategy enhances coverage and robustness. In terms of descriptor expression quality, both ORB and the proposed method achieve an information entropy metric of 0.98, indicating a balanced distribution of descriptor bits between 0 and 1, which demonstrates strong discriminative power. However, although SIFT and SURF descriptors have higher dimensionality and floating-point precision, their dimensional distribution on standard images is uneven. This results in normalized information entropies that are generally below 0.7. Furthermore, the descriptor bias of the proposed method is 0.49, which is comparable to the 0.51 bias of ORB. An ideal descriptor bias value approaches 0.5, which indicates a nearly uniform probability of each bit being 1 across a large sample. This metric shows that the constructed descriptor has good bit balance with large samples, which contributes to enhanced matching robustness. Regarding the matching repeat rate, the proposed method achieves a value of 0.85, which outperforms the values of 0.79 achieved by ORB and SURF and slightly exceeds the value of 0.83 achieved by SIFT. These results demonstrate superior stability and re-localization capability under scale and viewpoint variations. The proposed method has a runtime of 20.56 ms, which is slightly higher than ORB's 16.77 ms but significantly faster than SIFT's 103.29 ms and SURF's 31.15 ms. This performance satisfies the 30 Hz real-time processing requirement, demonstrating excellent engineering adaptability.

Further analysis of the stability and adaptability of each method in dynamic image sequence processing is shown in Fig. 9.



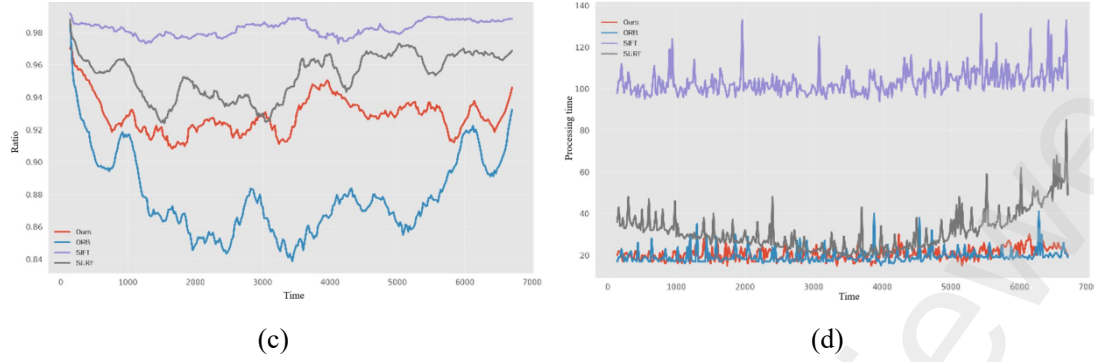


Fig. 9 Time-dependent Curves of Core Metrics: (a) Trend in the Number of Feature Points; (b) Trend in the Number of Valid Matches; (c) Trend in the In-point Rate; (d) Trend in Processing Time.

Fig. 9(a) shows the trend in the number of feature points across consecutive frames. SIFT maintains a count of around 500 points, while ORB and the proposed method exhibit minimal fluctuations, demonstrating effective control of the target quantity. In contrast, SURF drops below 300 points between frames 2,000 and 5,000, indicating that its corner detector struggles to adapt in low-texture regions. This leads to reduced matching stability. Fig. 9(b) shows that our method maintains the highest and most stable number of effective matches throughout most of the sequence. SIFT and ORB are slightly lower but steady. SURF, however, experiences multiple significant drops, reflecting its limited adaptability in complex, dynamic sequences. Fig. 9(c) displays the temporal trend of the in-point rate. SIFT maintains a high level of approximately 0.98, leveraging its geometric consistency advantage as a floating-point feature. SURF also sustains a relatively high level. Our method ranges from 0.92 to 0.95. This is slightly lower than floating-point features, but it significantly outperforms ORB while exhibiting superior temporal stability. Fig. 9(d) illustrates the computational time trends for the four methods. SIFT has high average processing times that frequently exceed 120 ms, making it unsuitable for real-time applications. SURF's processing time fluctuates significantly with the number of features but stabilizes near 60 ms in later iterations. ORB and the proposed method remain stable at approximately 20 ms, indicating that the proposed method is computationally efficient and enhances performance.

To analyze the stability and distribution characteristics of feature descriptions across methods, this paper performs a distribution analysis of descriptor construction results from two perspectives: descriptor bias and information entropy (Fig. 10). Fig. 10(a) shows the descriptor bias distribution with an ideal value of 0.5. The proposed method maintains a value of approximately 0.49 throughout the sequence, demonstrating minimal fluctuation, excellent structural symmetry, and statistical equilibrium. ORB, on the other hand, exhibits significant fluctuation, which could compromise matching robustness. Fig. 10(b) illustrates information entropy variation. The proposed method approaches the theoretical optimum of 1.0, which is comparable to ORB and indicates rich and balanced information. In contrast, SIFT and SURF demonstrate lower entropy values with substantial fluctuation, suggesting potential issues with feature redundancy or directional bias.

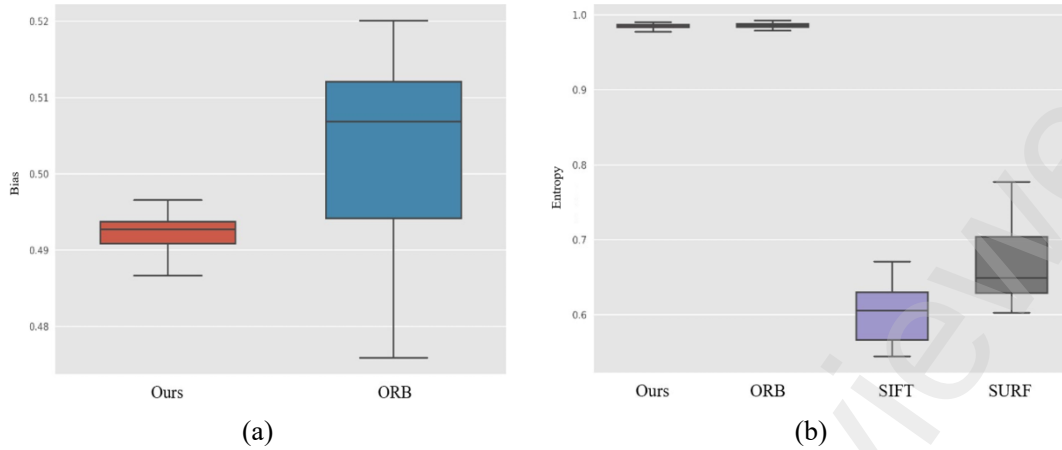


Fig. 10 Statistical Distribution of Feature Descriptors: (a) Distribution of Descriptor Bias; (b) Distribution of Descriptor Entropy Values

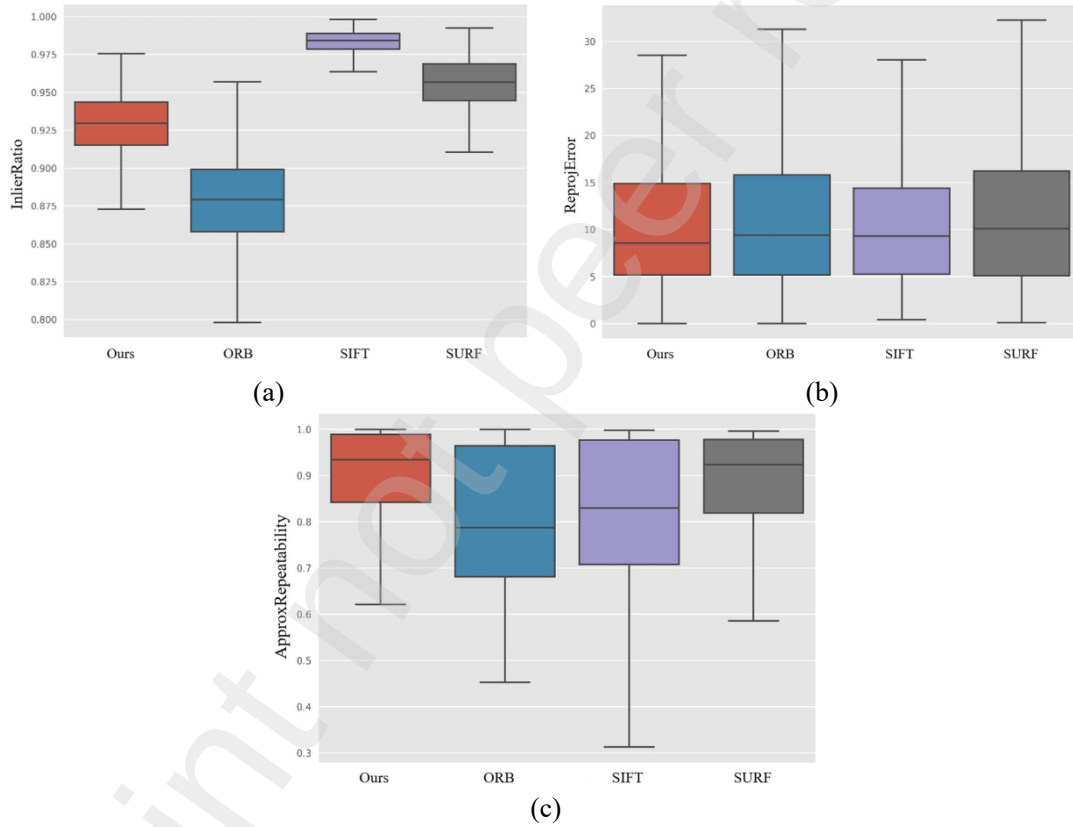
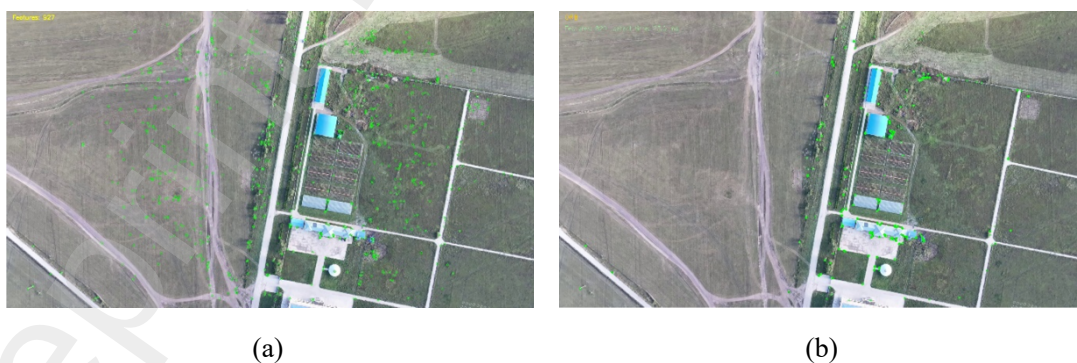


Fig. 11 Distribution of Matching Consistency and Repetition Rate Metrics: (a) Distribution of In-Point Rate; (b) Distribution of Repjection Error; (c) Distribution of Repetition Rate

To analyze the matching consistency and repositioning capability of each method under spatial geometric constraints, box plots of the sequence distributions for the in-point rate, reprojection error, and approximate repeat rate are presented, as shown in Figure 11. Fig. 11(a) displays the distribution of in-point rates after RANSAC geometric validation for different methods. SIFT and SURF maintain high in-point rates (mean>0.97) and low variance, indicating that floating-point descriptors achieve high accuracy in geometric model fitting. However, their computational overhead is substantial, making them difficult to apply on embedded platforms. In contrast, the proposed method

achieves a median in-point rate of approximately 0.93, outperforming ORB (0.88), with a tighter interquartile range, indicating more stable matching consistency across diverse scenes. This advantage stems from the direction-encoding enhancement and CLAHE normalization strategy, which effectively boost the structural matching robustness of the binary descriptor. Fig. 11(b) shows that the reprojection errors for single-objective matrix estimation across all methods are concentrated within the 5-15 pixel range. Our method exhibits a more compact interquartile range and fewer outliers, comparable to SIFT performance, indicating high stability and good accuracy in image-space structural fitting. In contrast, while ORB achieves similar average error, its outlier distribution is broader with greater geometric consistency fluctuations, reflecting error variability caused by directional mismatches. As shown in Fig. 11(c), our method outperforms ORB, SIFT, and SURF in both the median approximate repetition rate (approximately 0.94) and distribution compactness, indicating superior adaptability to scale and viewpoint variations. This performance enhancement stems from the optimized image-space equilibrium achieved by structured FAST feature extraction, coupled with the directional and local brightness normalization enhancements in the G^2 -ORB descriptor. These improvements enable stable and repeatable feature representation even in challenging scenarios such as blurring, texture degradation, or occlusion.

Fig. 12 shows that in a real-world scene containing roads, farmland, and low-texture grassland, the feature points extracted by our method are more evenly distributed. High density is notably maintained in the grassland, field ridges, and road intersection areas, which significantly enhances the response capability in weakly textured regions. This improvement stems from the locally adaptive threshold mechanism and the feature-density-driven quad-tree spatial balancing strategy. These strategies effectively mitigate the performance degradation of traditional methods in areas with varying textures. Other approaches extract abundant features in high-texture regions but suffer from severe deficiencies in low-texture areas, resulting in uneven distribution and strong structural dependency.



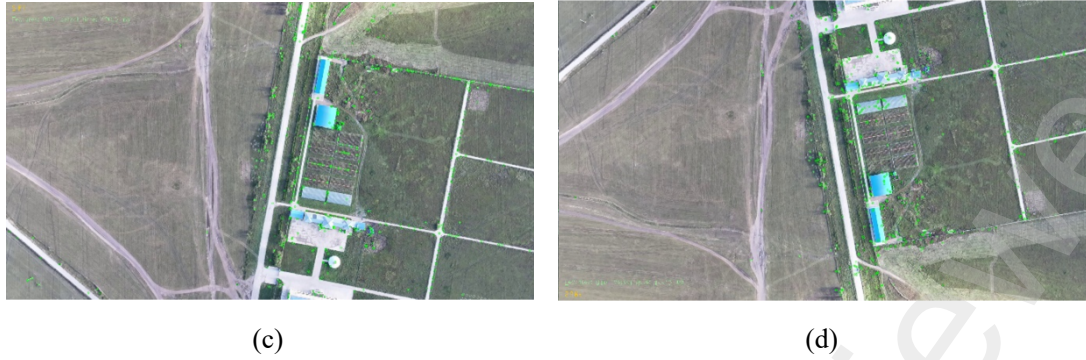


Fig. 12 Visualization of Spatial Distributions of Keypoints Extracted by Different Methods: (a) Distribution of Feature points Extracted by the Proposed Method; (b) Distribution of Feature Points Extracted by the ORB Method; (c) Distribution of Feature Points Extracted by the SIFT Method; (d) Distribution of Feature Points Extracted by the SURF Method.



Fig. 13 Dynamic Mesh Partitioning and Keypoint Distribution Maps: (a) Mesh Partitioning for Uneven Texture Distribution Frames; (b) Mesh Partitioning for Sparse Texture Distribution Frames.

Fig. 13 shows the boundary map of the regions after the dynamic grid was partitioned and merged during feature extraction. The distribution of the extracted keypoints is overlaid on the map. Red boxes indicate grid cells that were merged based on an assessment of texture complexity, and green dots represent the final extracted keypoints. The grid maintains high resolution along distinct boundaries of fields and buildings to ensure perception accuracy and feature detail. In grassy areas with weak texture, multiple grids merge into larger detection units, reducing redundant operations and improving processing efficiency. The feature points maintain an appropriate density, which indicates that the adaptive threshold mechanism effectively activates corner points in regions with weak texture. This improves coverage while preventing an extreme distribution.

5.2 HPatches Dataset Validation

To evaluate the performance of the proposed method further, experiments were conducted using the publicly available HPatches dataset. The results are presented in Table 3. The results demonstrate that the proposed method is highly robust and capable of accurately representing features while balancing computational efficiency and geometric consistency. In terms of total processing time, the proposed method averages 29.28 ms with a similar number of feature points. This is comparable to ORB (31.82 ms) and significantly faster than SURF (79.38 ms) and SIFT

(178.62 ms). Regarding the in-point rate and reprojection error, the proposed method achieves an average in-point rate of 94.3%. This rate outperforms those of ORB and SURF, approaching the rate of SIFT (95.7%). The reprojection error is 1.38 pixels, essentially matching SIFT's 1.33 pixels, demonstrating excellent geometric consistency and precision control. In terms of repeatability, our method (0.84) slightly trails SIFT (0.88) but matches ORB (0.83). Considering that our approach employs a binary descriptor and operates at a faster speed, these results demonstrate a significant advantage in balancing performance and efficiency.

Table 3 Average Performance Comparison of Methods on the HPatches Dataset

Method	Total duration (ms)	Number of feature points	Interior point matching rate (%)	Reprojection error (px)	Duplication rate (%)
SIFT	178.62	800.18	95.7	1.33	88
SURF	79.38	1194.48	93.6	1.61	87
ORB	31.82	799.98	90.2	1.40	83
Ours	29.28	862.36	94.3	1.38	84

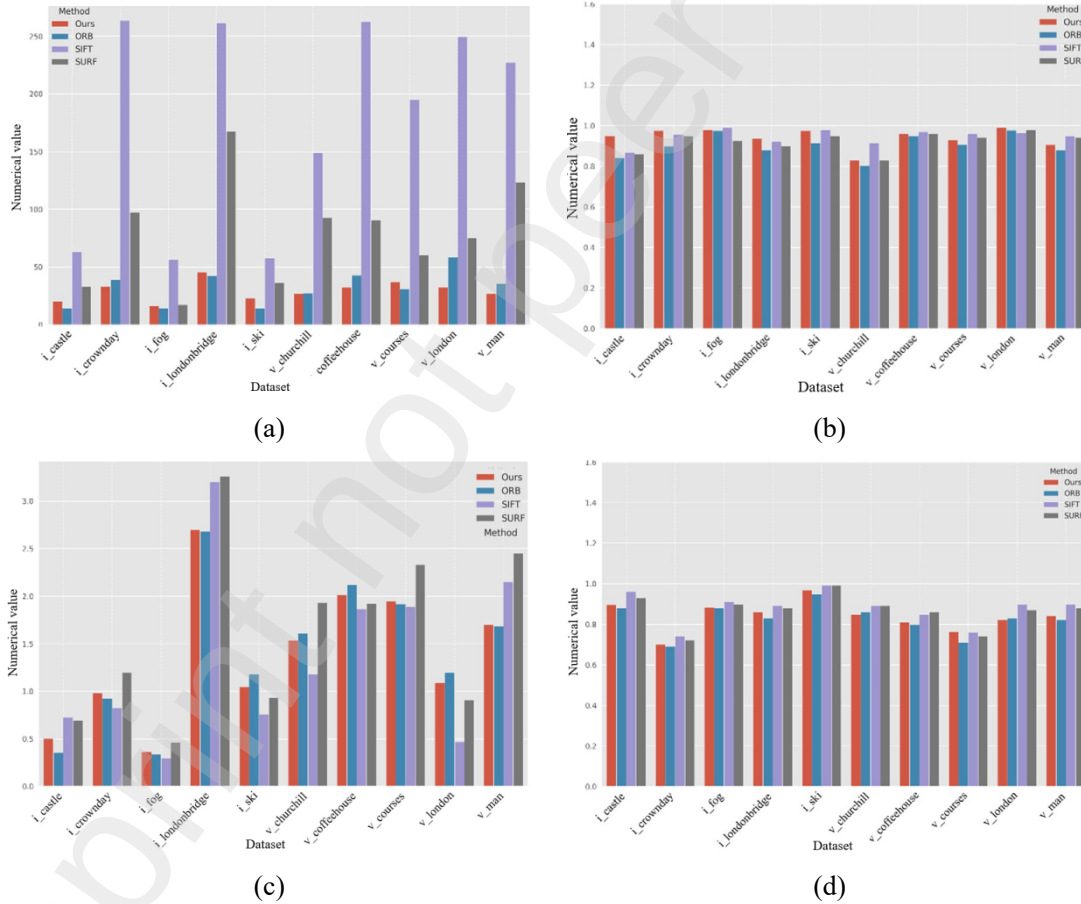
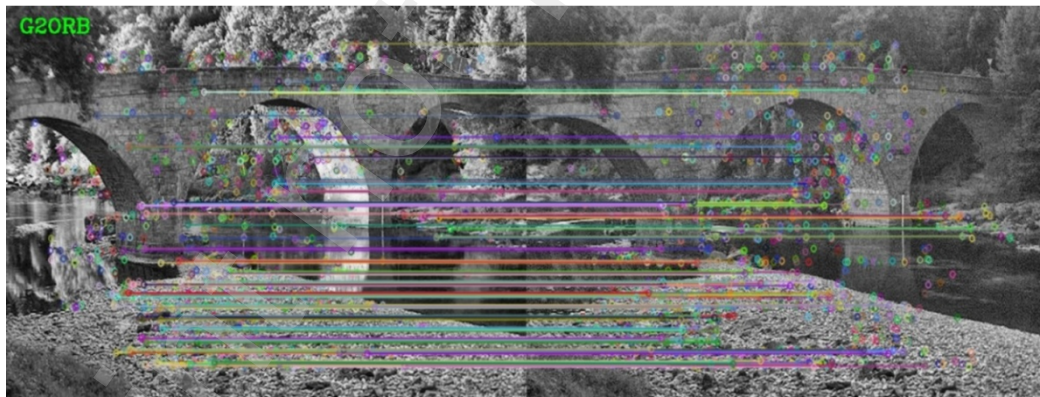


Fig. 14 Comparison of Group Metrics: (a) Runtime Comparison; (b) In-point Rate Comparison; (c) Reprojection Error comparison; (d) Repeatability Comparison

To analyze the adaptability and stability of the proposed algorithm under different types of visual disturbances, this paper divides the image groups into two subsets based on the HPatches dataset structure: lighting variation (i_ prefix) and viewpoint variation (v_ prefix). Fig. 14 shows

the bar charts plotting key metrics and performing group comparisons of ten image pairs.

Overall, SIFT was the slowest approach and lacked real-time capability across all subsets. In contrast, ORB and the proposed method maintained stable processing times below 30 ms, demonstrating superior lightweight performance. In terms of in-point rate, the proposed method demonstrates the most robust performance in the lighting transformation category. It outperforms ORB and SURF approaches, and even partially surpasses, SIFT. This advantage stems from the CLAHE lighting normalization processing introduced in the method and the directional enhancement mechanism, which effectively mitigate the interference of local brightness variations on feature matching accuracy. In the viewpoint variation group, SIFT maintains its lead due to its high-dimensional floating-point features. However, the proposed method follows closely behind, achieving a significantly higher in-point rate than ORB. This validates the adaptability of our directional encoding strategy to local structural variations. The proposed method demonstrates a low overall reprojection error level. Although SIFT has superior geometric fitting capabilities, it has a higher computational cost. ORB and SURF show relatively high errors in certain groups and significant fluctuations in geometric consistency, indicating limitations. In terms of repetition rates, our method outperformed ORB in the illumination variation category, suggesting that the ORB approach is more sensitive to brightness changes. In the viewpoint variation category, our method's repetition rate was slightly lower than SIFT's but higher than ORB's. This demonstrates the structure-direction enhancement mechanism's significant effectiveness in maintaining feature consistency across viewpoints.

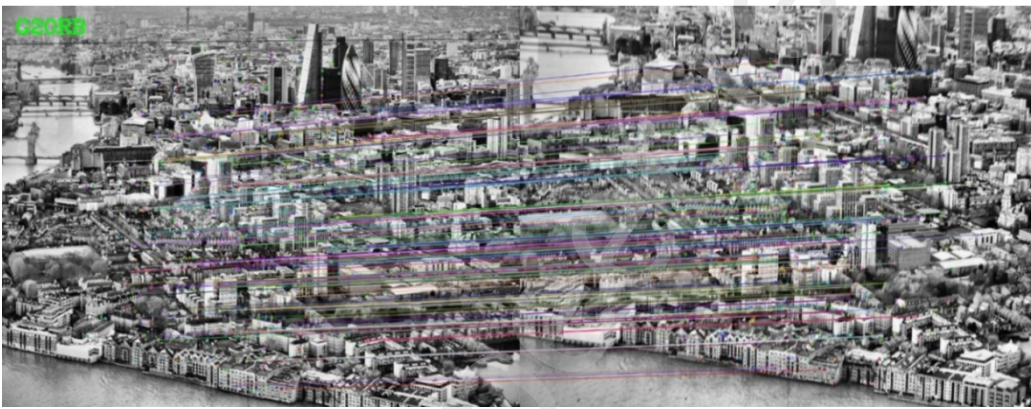


(a)



(b)

Fig. 15 Visualization of Illumination-adaptive Matching Results : (a) Visualization of The Proposed Method; (b) Visualization of the ORB Method;



(a)



(b)

Fig. 16 Visualization of Perspective-based Matching Results: (a) Visualization of the Proposed Method; (b) Visualization of the ORB Method;

A visual comparison analysis was conducted using two typical interference scenarios: lighting and viewpoint changes. Fig. 15 and 16 show the matching visualization results of the proposed method and the ORB method for two sets of images with lighting and viewpoint variations. Fig. 15 shows that, in the illumination variation group, our method produces more uniform matching lines with superior geometric consistency. Notably, our method maintains robust corner detection capability and matching consistency, even in the dark left region. This demonstrates greater

robustness than the ORB method. In contrast, ORB loses feature points in this area, resulting in sparse matching lines and poor geometric consistency. Fig. 16 shows that, in the bird's-eye city view image from the viewpoint variation group, both the proposed method and ORB stably extract keypoints along building edges and road contours, despite the rich image texture and dense building structures. The number of matched points, distribution range, and geometric structure of the connecting lines are similar, yielding stable matching results with a well-preserved geometric structure. These results indicate that, in high-texture scenes, our method maintains feature stability and geometric adaptability comparable to traditional methods, balancing computational efficiency and demonstrating good universality and versatility.

5.3 EuRoC Dataset Validation

This paper uses the Evo trajectory error analysis tool to evaluate the pose trajectories generated by different methods in a quantitative way. All methods use the same initial pose and undergo SE(3) registration. The deviation between the trajectories and the ground truth is primarily measured by the absolute pose error (APE), with the following metrics being statistically calculated: maximum, minimum, mean, median, standard deviation and root mean square error (RMSE). Evaluations were conducted on five sequences (MH01-05) categorised as easy, medium or difficult based on flight speed and texture richness. These correspond to low-dynamic, moderately challenging and high-challenge scenarios, respectively. The results for each metric are presented in Table 4.

Table 4 Comparison of Error Metrics for Different Methods

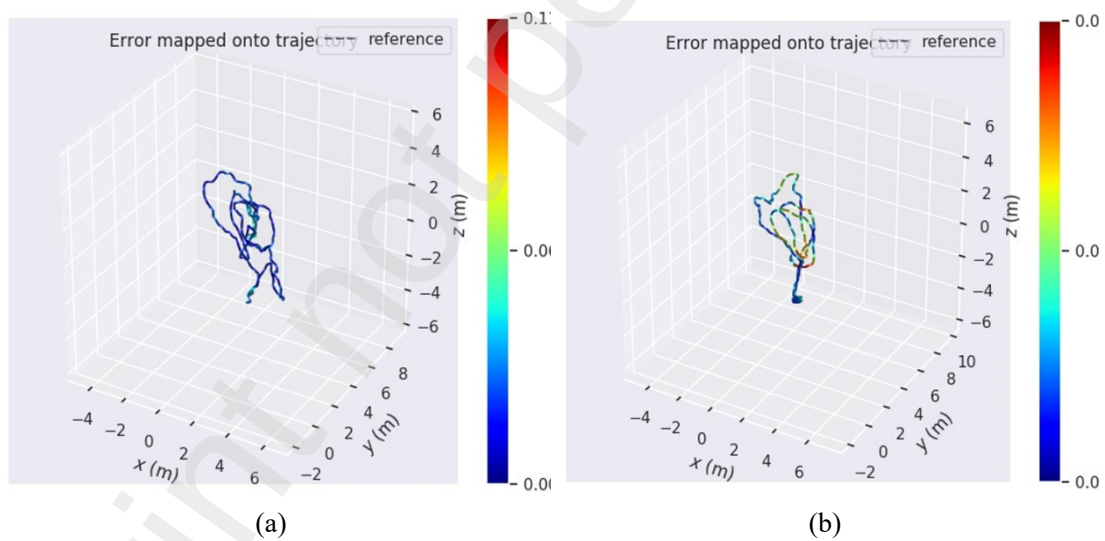
Dataset	Method	Maximum/m	Mean /m	Median/m	Minimum/m	SD/m	RMSE/m
MH_01_easy	VINS	0.471	0.149	0.128	0.023	0.088	0.173
	ORB3	0.088	0.031	0.025	0.002	0.021	0.037
	Ours	0.118	0.023	0.021	0.002	0.011	0.025
MH_02_easy	VINS	0.206	0.071	0.060	0.024	0.040	0.082
	ORB3	0.093	0.031	0.027	0.002	0.016	0.035
	Ours	0.058	0.021	0.017	0.001	0.011	0.024
MH_03 _medium	VINS	0.268	0.143	0.146	0.031	0.045	0.150
	ORB3	0.113	0.034	0.030	0.002	0.018	0.038
	Ours	0.125	0.028	0.028	0.002	0.015	0.031
MH_04 _difficult	VINS	0.285	0.183	0.174	0.082	0.041	0.187
	ORB3	0.337	0.051	0.042	0.005	0.032	0.061
	Ours	0.171	0.041	0.034	0.004	0.023	0.047
MH_05 _difficult	VINS	0.498	0.291	0.272	0.129	0.086	0.303
	ORB3	0.198	0.043	0.038	0.004	0.021	0.048
	Ours	0.127	0.040	0.035	0.002	0.020	0.044



Fig. 17 Bar Chart Comparison of Error Metrics: (a) Error Metrics for MH01 Sequence; (b) Error Metrics for MH02 Sequence; (c) Error Metrics for MH03 Sequence; (d) Error Metrics for MH04 Sequence; (e) Error Metrics for MH05 Sequence.

To provide a more intuitive comparison of statistical metrics across different algorithms, this paper further plots error bar charts corresponding to Table 4, illustrating the variation in metric values across the five sequences, as shown in Fig. 17. The experimental results show that VINS-Fusion has a significantly higher maximum error and root mean square error (RMSE) than other methods across multiple sequences. In highly dynamic scenarios such as MH_05_difficult, for example, its RMSE reaches 0.303 m, indicating pronounced error accumulation under strong disturbance conditions. The ORB-SLAM3 method exhibited relatively stable performance in low-

to-medium dynamic scenarios, with respective RMSEs of 0.037 m and 0.035 m. However, error fluctuations increased significantly in more complex environments. For example, the maximum error reached 0.337 m in the MH_04_difficult sequence, suggesting that it is not well-suited to highly disturbed scenarios. By contrast, the proposed method demonstrates superior overall localisation accuracy and stability of error across all sequences. Across the five sequences ranging from MH_01_easy to MH_05_difficult, the RMSE of the proposed method remained consistently below 0.047 m, with an average overall RMSE of 0.0342 m. This represents a reduction in error of approximately 80.9% compared to VINS-Fusion (average 0.179 m) and 21.9% compared to ORB-SLAM3 (average 0.0438 m). It demonstrated robust stability and adaptability, particularly in medium-to-high dynamic sequences such as MH_03_medium and MH_04_difficult. Furthermore, in terms of standard deviation, the proposed method exhibited the lowest error variability across all scenarios, with standard deviations (STDs) consistently below 0.020 m. This significantly outperforms the averages of 0.0216 m and 0.021 m for ORB3 and VINS, respectively. Our method exhibited the lowest error variability across all scenarios, with STDs consistently below 0.020 m. This significantly outperforms the averages of 0.0216m and 0.060m for ORB3 and VINS respectively, demonstrating superior overall accuracy and trajectory estimation stability. This validates the effective compensation provided by UWB ranging constraints for state estimation under conditions of weak texture and high dynamic interference.



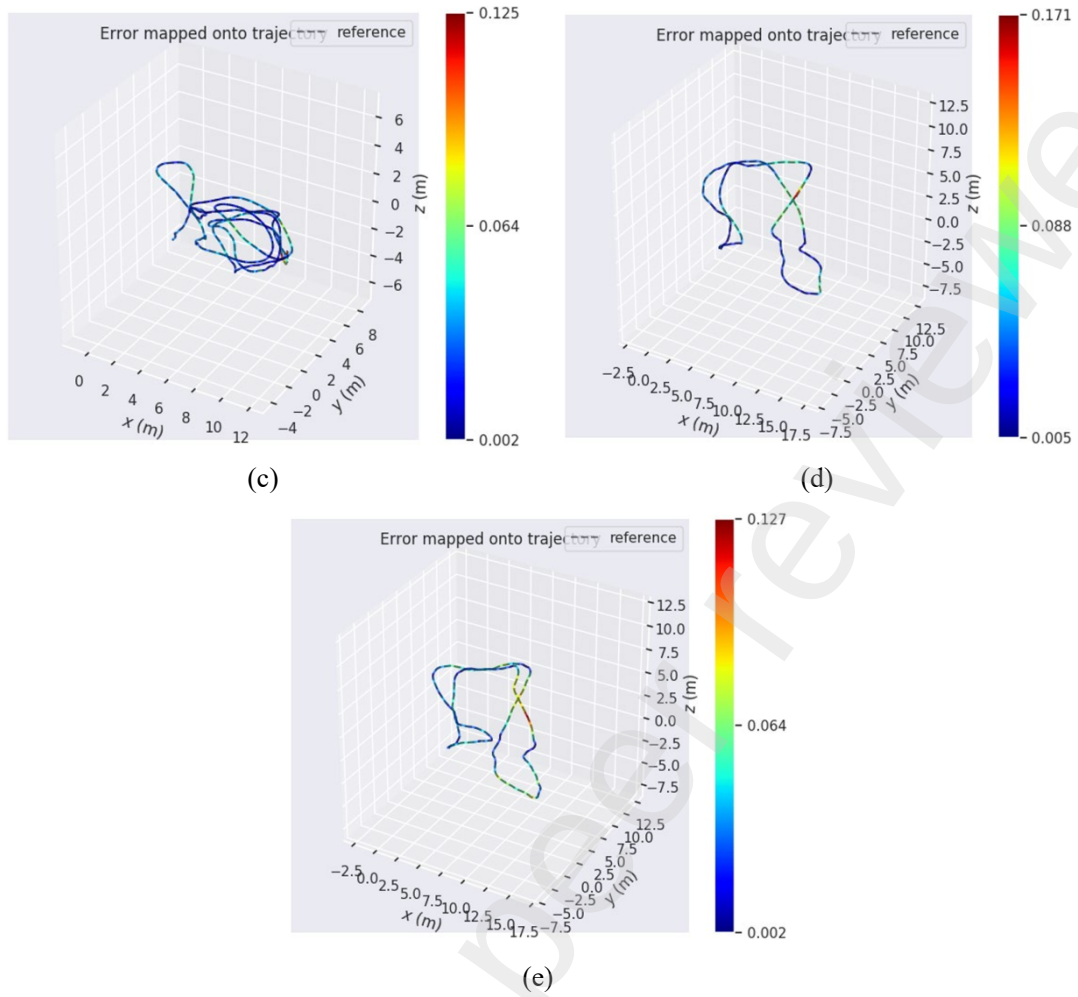
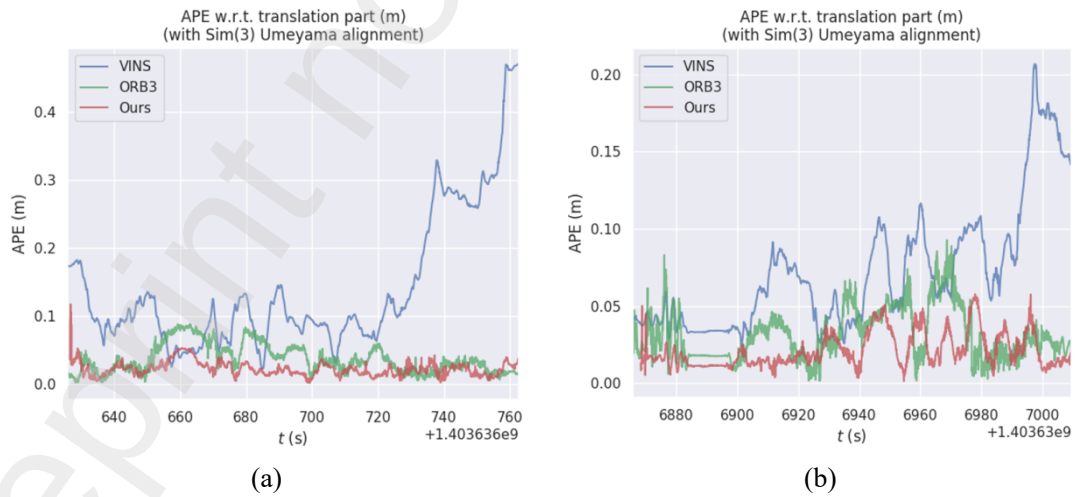


Fig. 18 Absolute Trajectory Error Plots: (a) MH01 Sequence Trajectory; (b) MH02 Sequence Trajectory; (c) MH03 Sequence Trajectory; (d) MH04 Sequence Trajectory; (e) MH05 Sequence Trajectory



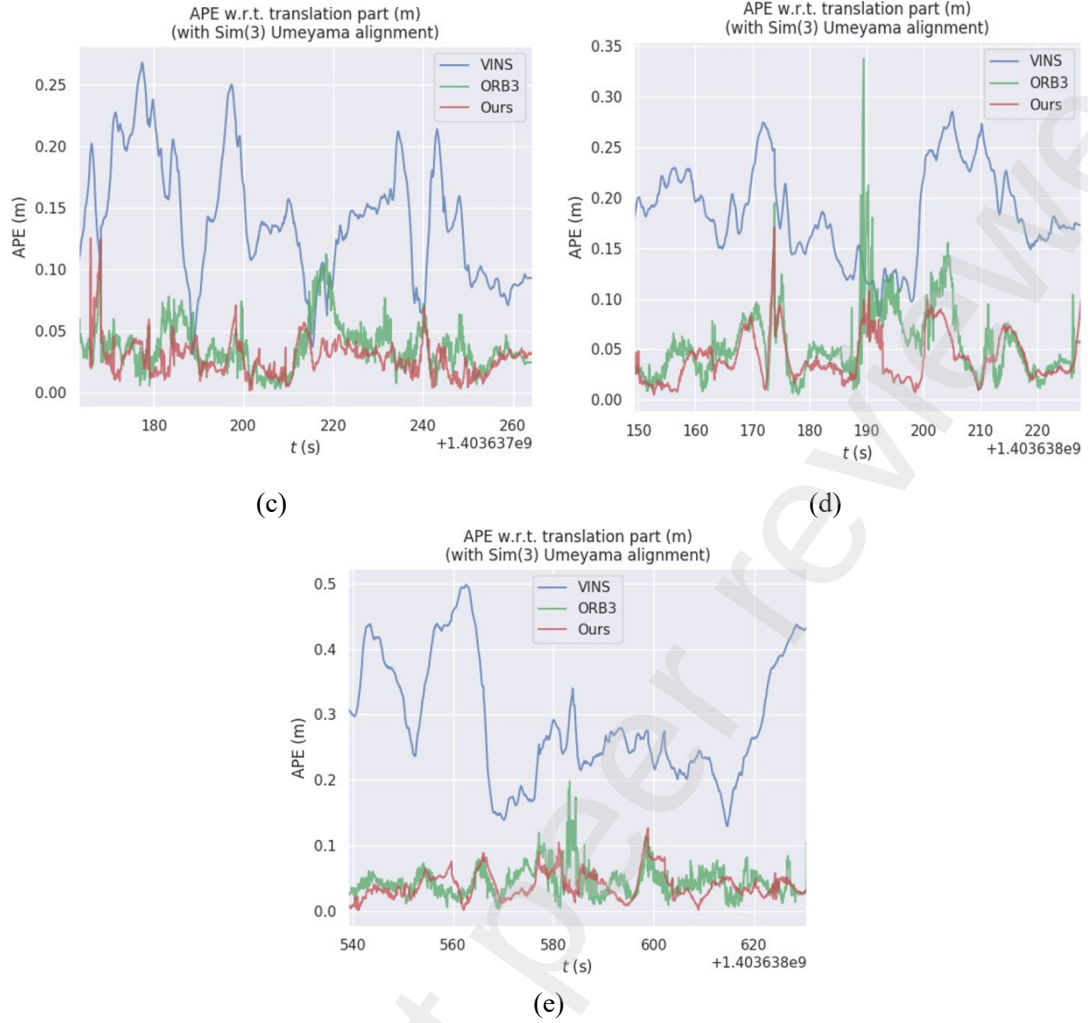


Fig. 19 Absolute Pose Error Curves: (a) Absolute Pose Error of MH01 Sequence; (b) Absolute Pose Error of MH02 Sequence; (c) Absolute Pose Error of MH03 Sequence; (d) Absolute Pose Error of MH04 Sequence; (e) Absolute pose error of MH05 sequence.

Fig.18 illustrates the visual distribution of errors between the estimated trajectory and the true trajectory, with color gradients representing the magnitude of positioning errors at different locations. In the MH_03_medium sequence, the system maintains good accuracy consistency in most trajectory bends as flight speed increases and texture complexity grows. Errors are primarily concentrated at a few turning points, with colour distribution predominantly in the green range and maximum error controlled to within 0.12 m. In the high-dynamic, low-texture MH_04_difficult and MH_05_difficult sequences, the system experiences rapid acceleration and deceleration, as well as sharp turns across multiple trajectory segments. Conventional visual-inertial methods are unstable in such scenarios. Visualisation shows that, despite minor error spikes in isolated segments, the overall error remains below 0.17 m. Trajectories closely match the ground truth and there is a sparse distribution of high-error red zones. This confirms that introducing UWB fusion constraints effectively mitigates errors under strong disturbances. The overall error remains below 0.17 m. The trajectory closely aligns with the ground truth, with a sparse distribution of high-error red zones. This validates the effectiveness of the UWB fusion constraint introduced in this paper in maintaining

representative UAV flight trajectories were collected for system validation. The UAV's high thrust-to-weight ratio and rapid attitude response simulate typical dynamic states. The first trajectory involved a high-speed takeoff and attack penetration featuring steep acceleration, climbs, and dives to examine the system's stability and transient error control under intense motion. The second trajectory consisted of low-altitude, stable cruising with gradual attitude changes to verify accuracy consistency and cumulative error control. Table 5 summarizes the error statistics for these two tasks and demonstrates the system's robust accuracy and stability under diverse dynamic conditions.

Table 5 Positioning Error Metric Results

Trajectory	Max/m	Mean/m	Median /m	Min/m	SD/m	RMSE/m
Trajectory 1	0.902	0.459	0.443	0.167	0.140	0.480
Trajectory 2	0.552	0.261	0.263	0.050	0.116	0.286

The results show that for trajectory 1 corresponding to the high-speed takeoff-attack penetration mission, the maximum error reached 0.902m, the average error was 0.459m, the RMSE reached 0.480m, and the standard deviation was 0.140m. This outcome indicates that despite short-term accuracy fluctuations under conditions of abrupt attitude changes and high acceleration, the system exhibits no significant cumulative error. Its positioning performance remains within acceptable limits, demonstrating a degree of dynamic robustness. For Trajectory 2, corresponding to low-altitude cruise missions, the system's positioning accuracy was more stable, with significantly improved error metrics compared to Trajectory 1: the mean error and RMSE were 0.261 m and 0.286 m, respectively, the standard deviation was controlled within 0.116 m, and the maximum error was only 0.552 m. These results fully demonstrate that the proposed method exhibits stronger error suppression capabilities and higher accuracy consistency under steady-state operating conditions.

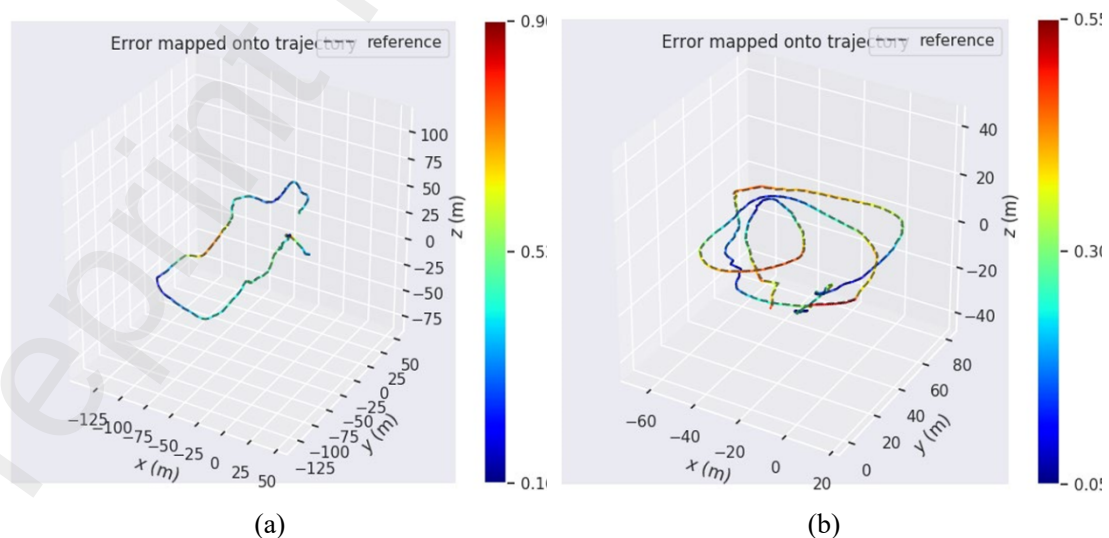


Fig. 20 Experimental Trajectory Diagrams: (a) Error Mapping Diagram for Trajectory 1; (b)

Error Mapping Diagram for Trajectory 2

Fig. 20 shows the distribution of errors between the estimated and true trajectories during the experiment. The system achieved sub-meter positioning accuracy overall under typical flight mission scenarios, demonstrating controllable error ceilings and rapid convergence capabilities under high dynamic disturbances. Fig. 20(a) shows that the overall trajectory trend is highly consistent with the ground truth. However, localized error peaks occur during climb and dive turns, with a maximum error of 0.902 m. During relatively stable attitude segments, errors rapidly converged, maintaining stable tracking overall. In the low-altitude cruise mission depicted in Fig. 20(b), the positioning accuracy became more uniformly concentrated, with enhanced spatial consistency and errors primarily clustered between 0.05 m and 0.3 m.

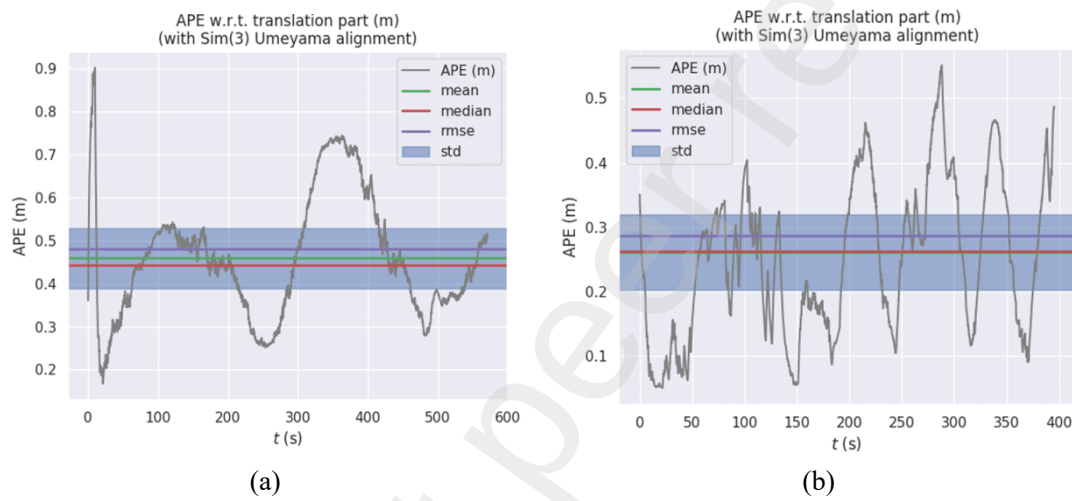


Fig. 21 Absolute Pose Error Diagram: (a) Absolute Pose Error for Trajectory 1; (b) Absolute Pose Error for Trajectory 2

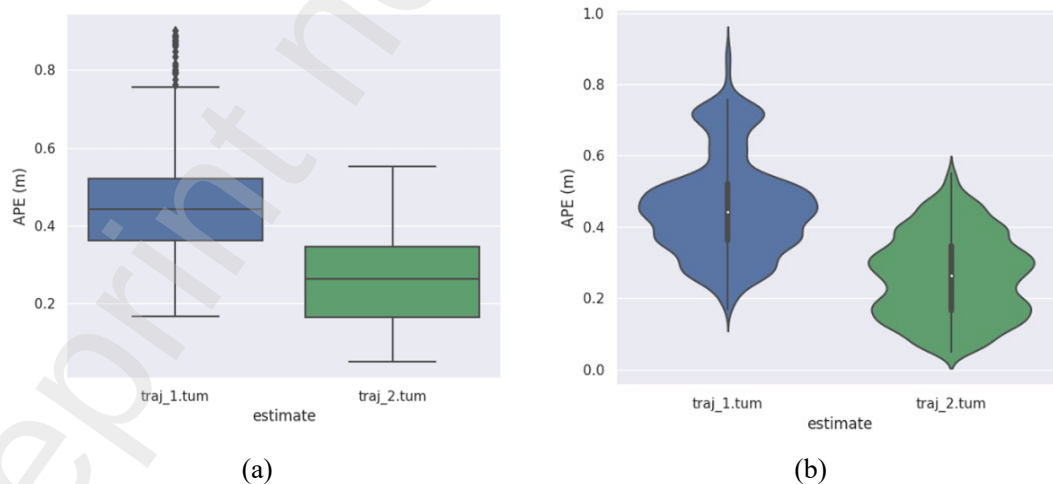


Fig. 22 APE Analysis Plots: (a) Trajectory APE Box Plot; (b) Trajectory APE Violin Plot

Fig. 21 shows the absolute position error (APE) curves of two trajectories over time. Fig 21(a) exhibits a significant error spike during the initial flight phase, where the positioning error temporarily surged to approximately 0.9m. However, it rapidly converged to the 0.4 – 0.6m range

within a very short time and maintained a relatively stable error level throughout the subsequent flight. This phase corresponds to the high-dynamic operating conditions triggered by strong excitation during the drone's takeoff process. This error response characteristic demonstrates that the proposed UWB-IMU initial state estimation strategy exhibits excellent robustness and rapid convergence capability. The system achieved initial convergence within seconds after takeoff and maintained stable tracking, demonstrating the strategy's ability to effectively compensate for information gaps caused by unobservable features or blurred images in the visual system during this phase. In contrast, methods based on visual/visual-inertial initialization, such as VINS-Fusion and ORB-SLAM3, frequently encountered failures in visual feature extraction and interruptions in tracking initialization under this scenario. This led to incomplete trajectory estimation and rapid error divergence. This further demonstrates the enhanced usability of the proposed fusion strategy during complex dynamic takeoff phases. Fig. 21(b) shows a smoother error curve for Trajectory 2, with subsequent errors fluctuating within 0.1-0.4m without significant divergence. Although minor local oscillations occurred in the middle-to-late stages, overall convergence and consistency remained robust. This indicates that the system exhibits minimal error fluctuations during steady-state cruise tasks, yielding more reliable estimation results.

Fig. 22(a) shows the APE box plots for two flight trajectories. Trajectory 1 has a median error of approximately 0.44 m and a wide interquartile range of errors. The maximum error approaches 0.8 m, and there are numerous outliers. These results suggest significant error fluctuations, frequent sporadic deviations, and relatively unstable robustness performance under high-dynamic conditions. In contrast, Trajectory 2's box plot is more compact, with a median error of about 0.26 m, few outliers, and a maximum error of 0.55 m or less. This reflects the system's higher precision, consistency, and error control capability under stable, low-dynamic flight conditions. Fig. 22(b) further illustrates the distribution pattern of trajectory estimation errors using a violin plot. Trajectory 1 has a broad contour with a bimodal structure, featuring a primary peak in the 0.4 – 0.5 m range and a long tail extending into the 0.6 – 0.8 m high-error interval. This pattern indicates significant error fluctuations under high-dynamic conditions and a notable proportion of high-bias regions in the estimation results. This morphology aligns with the outlier phenomenon observed in the preceding box plot. In contrast, Trajectory 2 exhibits a more concentrated and symmetrical error density distribution. Its primary error peak occurs within the 0.2 – 0.3 m range, with rapidly decaying tails and no discernible trend toward high-bias expansion. This suggests more reliable estimation accuracy during stable cruising phases, which are characterized by well-centered and consistent errors.

In summary, the two representative trajectories demonstrate the multi-source fusion positioning system's accuracy and stability under different flight conditions. In contrast, VINS-Fusion and ORB-SLAM3 tend to diverge during high-dynamic takeoff phases due to initialization failures. This underscores the practicality of the UWB-IMU combined initialization strategy

presented in this paper, which furnishes robust initial values regardless of visual constraints. This strategy ensures the convergence and accuracy of map optimization, demonstrating the system's advantages for deployment in high-dynamic environments.

6 Conclusion

(1) A multi-source fusion positioning method integrating IMU, visual modules, and UWB ranging modules is proposed. This approach fully leverages the complementary advantages of the three sensor types across temporal-frequency, error characteristics, and spatial constraint dimensions: IMUs offer high frequency and excellent short-term stability; visual modules possess strong environmental perception capabilities and are well-suited for relative motion observation; UWB signals feature low energy density, broad spectrum coverage, strong anti-interference capabilities, and can provide global high-precision positioning constraints.

(2) A structured visual feature optimization method is proposed. By employing a locally texture-aware adaptive FAST detection mechanism and a density-driven spatial balancing strategy, while integrating gradient and grayscale information in descriptor construction, CLAHE enhancement and Top-K direction sparse coding are introduced. This enhances the spatial distribution balance and directional robustness of feature points, thereby improving the accuracy and stability of visual observations during UAV positioning.

(3) A graph optimization positioning method integrating IMU/vision/UWB is proposed. An UWB-IMU initial state estimation strategy based on extended Kalman filtering is adopted, enhancing the system's convergence speed and accuracy during high-dynamic initial phases. Within the graph optimization framework, a UWB differential ranging residual factor and an inter-frame state compensation mechanism based on IMU pre-integration were designed to enhance the system's global constraint capability and observation time synchronization. Adaptive sliding window scheduling further improves real-time performance and optimization efficiency. Unified modeling and tightly coupled joint estimation of multi-source information were achieved, effectively boosting the positioning performance of UAVs in anti-access environments.

CRedit authorship contribution statement

Jia Lu: Writing—original draft, Formal analysis. Zuyin Zhang: Validation. Yishen Qi: Investigation, Software. Ping Song: Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The authors do not have permission to share data.

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